# RV UNIVERSITY

School of Computer Science & Engineering · B.Tech (Hons) · Semester V · Academic Year 2026-27

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| **Course Code** | CS310 | **Course Title** | Digital Image Processing |
| **L : T : P** | 3 : 0 : 1 | **Credits** | 4 |
| **CIE : SEE** | 50 : 50 | **Prerequisite** | Linear Algebra, Programming in Python |

## COURSE OUTCOMES

| CO | On completion of the course, students will be able to |
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| CO1 | Explain how digital images are acquired, sampled, quantised and represented in different colour spaces. |
| CO2 | Apply spatial and frequency-domain filters to enhance and restore images. |
| CO3 | Analyse edges, regions and morphology to segment objects in an image. |
| CO4 | Design a small image-processing pipeline including compression, using OpenCV. |

## COURSE CONTENT

**Unit 1 – Digital Image Fundamentals (9 hours)**
Image acquisition with a camera sensor; exposure, ISO and sensor noise; sampling and quantisation; spatial and intensity resolution; pixel neighbourhoods and connectivity; colour models – RGB, HSV and YCbCr; gamma correction.

**Unit 2 – Image Enhancement (10 hours)**
Intensity transformations – negative, log and power-law; histogram processing and histogram equalisation; spatial filtering and convolution; smoothing with box and Gaussian filters; sharpening with the Laplacian and unsharp masking; the 2-D Fourier transform and filtering in the frequency domain – ideal, Butterworth and Gaussian low-pass and high-pass filters.

**Unit 3 – Restoration and Segmentation (10 hours)**
Noise models – Gaussian, salt-and-pepper; restoration with mean and median filters; edge detection – gradient, Sobel, Prewitt and Canny; thresholding and Otsu's method; region growing and region splitting/merging; morphological operations – erosion, dilation, opening and closing.

**Unit 4 – Compression and Applications (10 hours)**
Redundancy in images; run-length and Huffman coding; the discrete cosine transform and JPEG compression; depth of field and defocus blur in images; image-processing pipelines for document scanning and medical images.

## LABORATORY EXPERIMENTS (P)

1. Read, display and convert images between RGB, HSV and grayscale in OpenCV; measure sensor noise at different ISO values.
2. Implement histogram equalisation and compare with contrast stretching.
3. Apply Gaussian, median and Laplacian filters; compare spatial and frequency-domain filtering.
4. Detect edges with Sobel and Canny and segment objects with Otsu thresholding.
5. Build a document-scanner pipeline: edge detection, perspective correction and JPEG compression.

**TEXT BOOKS**

1. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed., Pearson, 2018.
2. A. K. Jain, *Fundamentals of Digital Image Processing*, PHI, 2015.